

Z-Curve for Face-Selectivity in FFA

Team: Z-Score Zombies

Member: Eyas Ayeshe, Junxia Wang, Mainak Deb, Shuheng Jiang, Zhimin Hu

What is the FFA?

Kanwisher, McDermott & Chun, 1997

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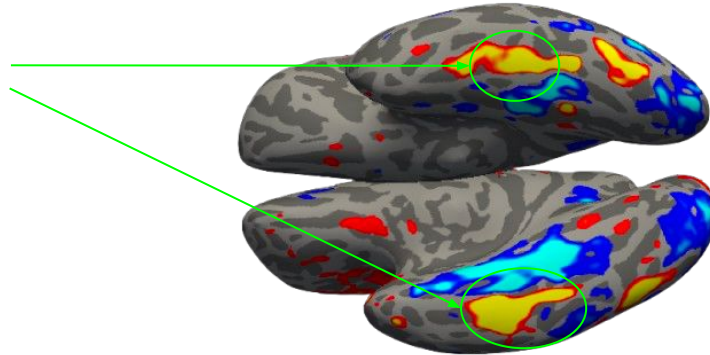
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- Fusiform Gyrus



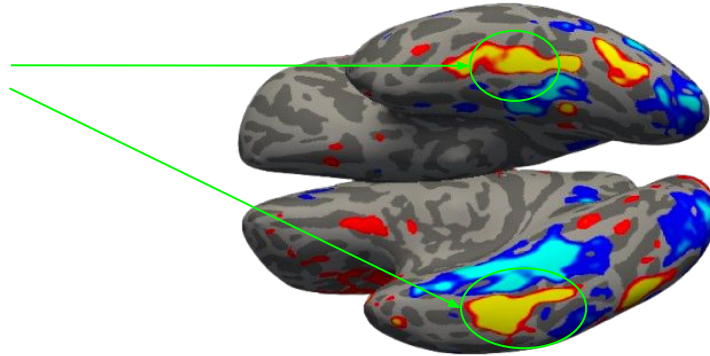
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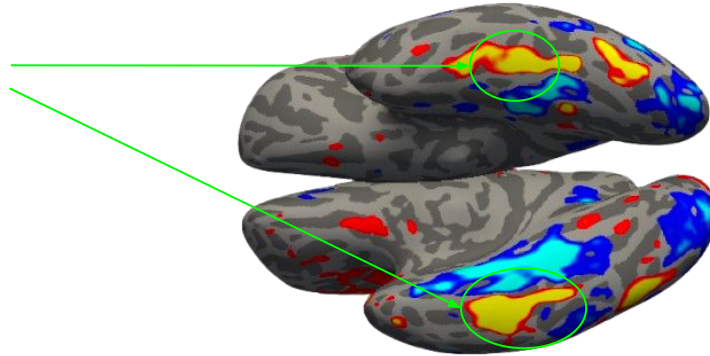
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How do we locate it in the brain?

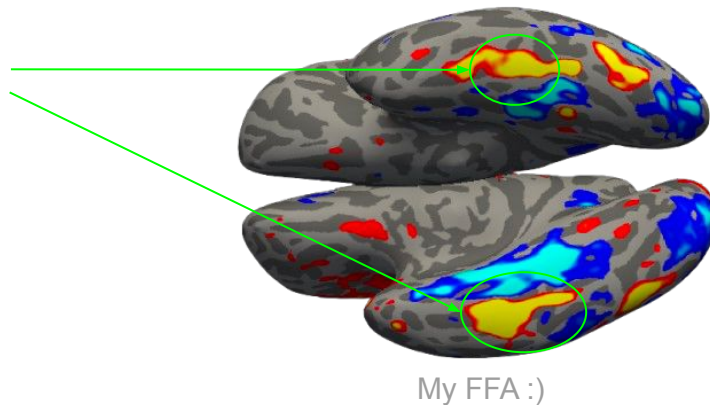
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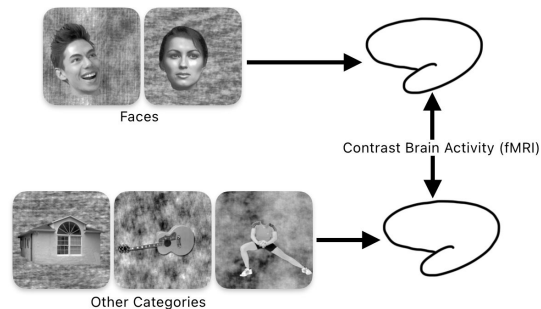
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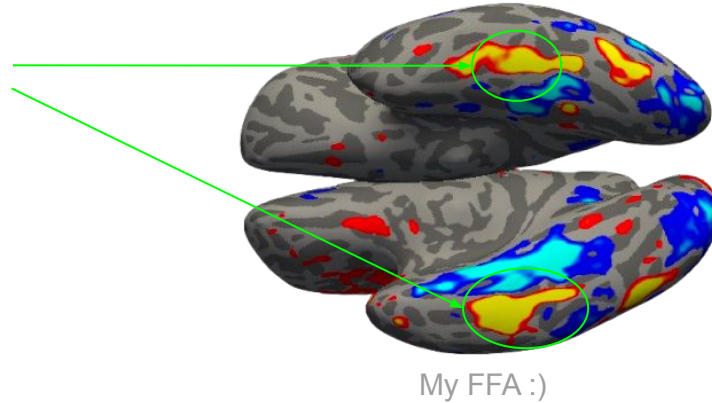
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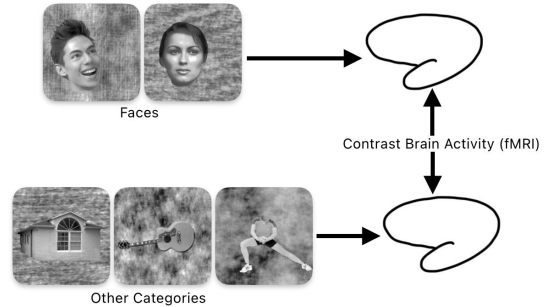
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Question

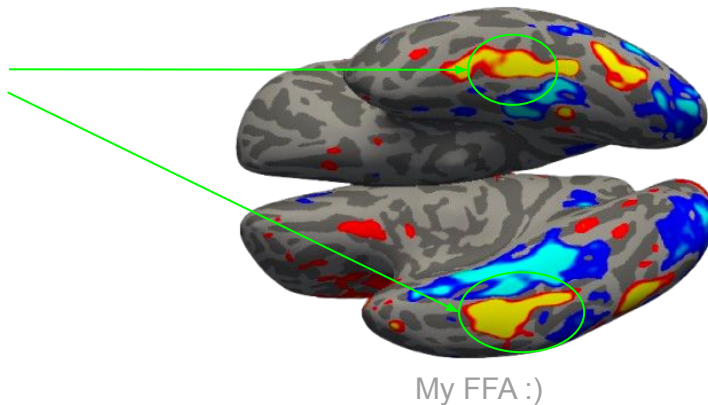
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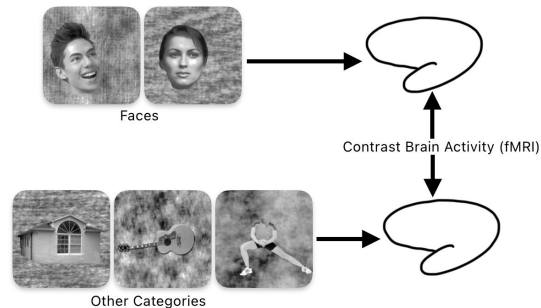
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Question

Does FFA face-selectivity reflect real brain specialization, or just publication bias?

Research Question and Hypothesis

- Does the published fMRI literature on FFA face-selectivity reflect genuine evidential value, or is it driven by selective reporting and statistical noise?
- We perform Z-curve analysis on the **Faces vs. Non-Faces** contrast in the FFA across ~30 studies
- We predict:
 - The **Estimated Discovery Rate (EDR)** will exceed the nominal α level (20-30%)
 - The **Expected Replication Rate (ERR)** will be substantially above zero
- A low EDR (~5%) paired with low ERR would indicate the significant literature **lacks robust evidential value**



What literature we looked into: Inclusion-Exclusion Criteria

Category	Inclusion Criteria	Exclusion Criteria
Methods	Human fMRI (BOLD response)	Non-fMRI (EEG, MEG, PET, NIRS); Non-human.
Design	Task-based face vs. non-face contrast	Resting state; Connectivity only
Analysis	Univariate activation in FFA	MVPA only; Correlation-only studies
Statistics	Exact p-value or sufficient statistics	Ambiguous reporting
Scope	Normative populations	Abnormal/Clinical populations

Decision Rules:

- **FFA Localization:** Functional or anatomical.
- **Laterality:** Default to left-lateralized if bilateral is present.
- **Multiple Tests:** Prioritize the first reported confirmatory comparison.

Example extraction table

Hypothesis	Pre-registered?	Design	IV	DV	Statistics Type	Reported p-value	Exact p-value
FFA is a domain-specific area, rather than an area that deals with individual differences or general expert recognition.	No	Within-subject	faces vs objects	Percent BOLD signal change in FFA	t-test	$p < .0001$	$p = .0001$
The FFA responds more strongly to faces than non-face objects.	No	Within-subject	Faces vs Objects, Scrambled faces, houses, hands	Percent BOLD signal change in FFA	F-test	$p < .01$	$p = .01$
FFA responds more strongly to faces than houses.	No	Within-subject	Faces vs Houses	Percent BOLD signal change in FFA	t-test	$p < .001$	$p = .001$

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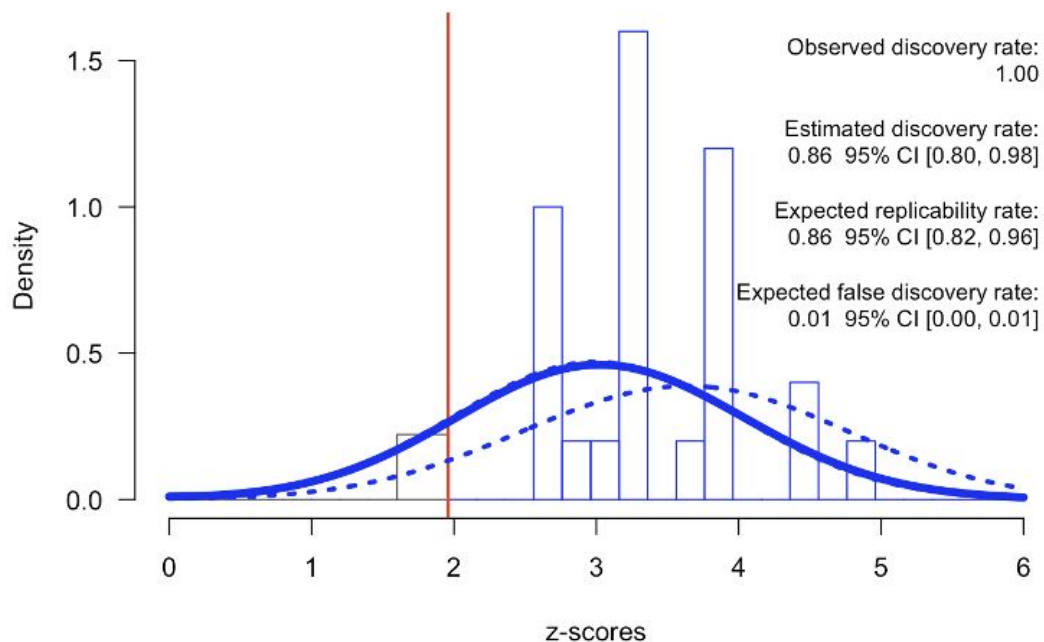
Excluded Literatures

paper	link	Hypothesis	Pre-registered?	Design	Independent Variable	Dependent Variable
Face-Specific Processing in the human fusiform gyrus	edu/jocn/article/9/5/605/3380/Face-Specific-Processing-in-the-Human-Fusiform?guestAccessKey=	Is the FFA revealed when the object-recognition system is shown general objects	No		faces vs objects	Voxel count
The challenge of localizing the anterior temporal face area: A possible solution	https://www.sciencedirect.com/science/article/pii/S1053811913005090	Coronal slice orientation improves localization of the ATL face area relative to standard axial orientation	No		NA	NA
Face Configuration Processing in the Human Brain: The Role of Symmetry	https://academic.oup.com/cercor/article/17/6/1423/421531?guestAccessKey=	The upright versus inverted faces (with the same image symmetry but unfamiliar object identity) showed robust differential activation in the FFA.	No		frontal-view faces vs. Fourier-amplitude-matched asymmetric phase-scrambled images	BOLD signal

- Starting pool of 30 articles two primary coders agreed on 25
- The five disagreements were almost all about ambiguous FFA reporting.
- Third coder to flag articles that didn't meet our codebook's clarity requirements
- Our final analysis sample has 27 z-values

Results and conclusion: Z-curve

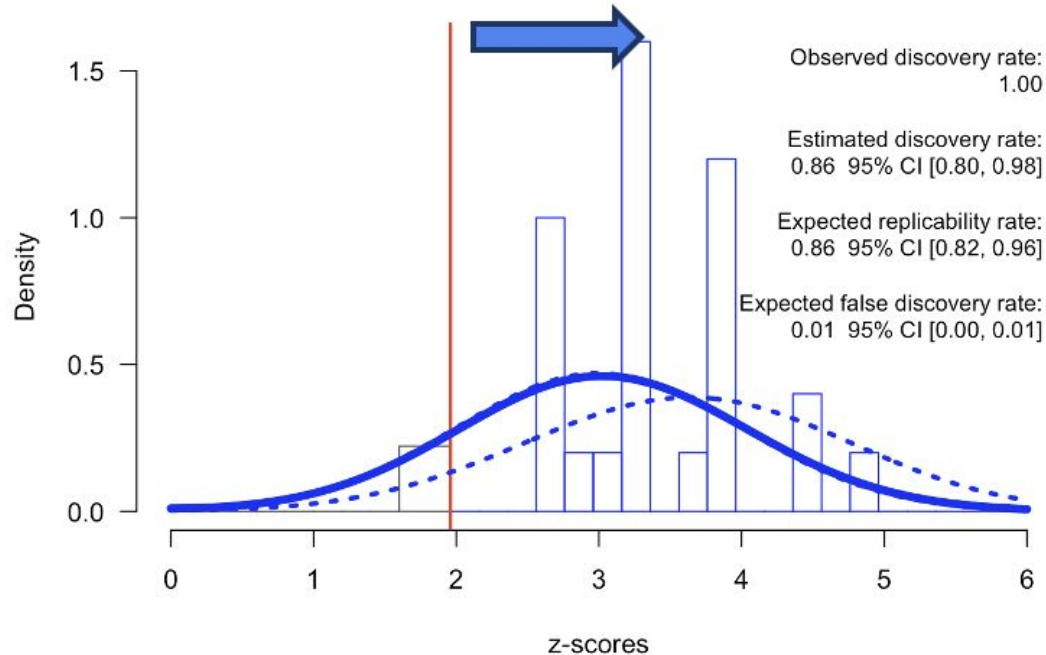
Z-curve from uploaded FFA extraction sheet



Z-curve fit to our 27 z-values

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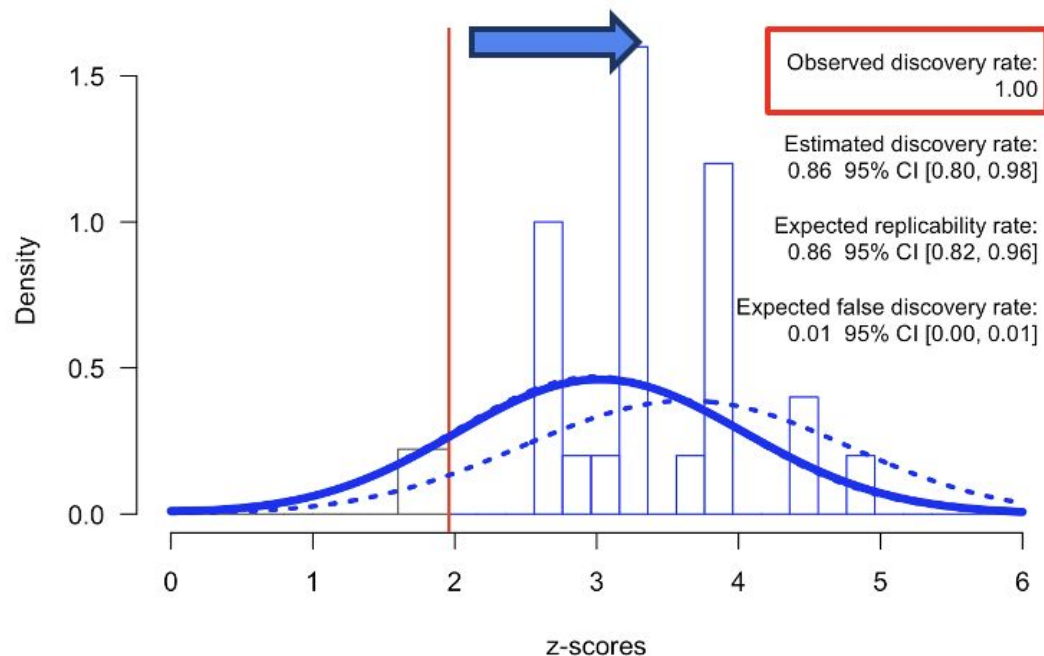
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Red Vertical Line $\rightarrow z = 1.96$

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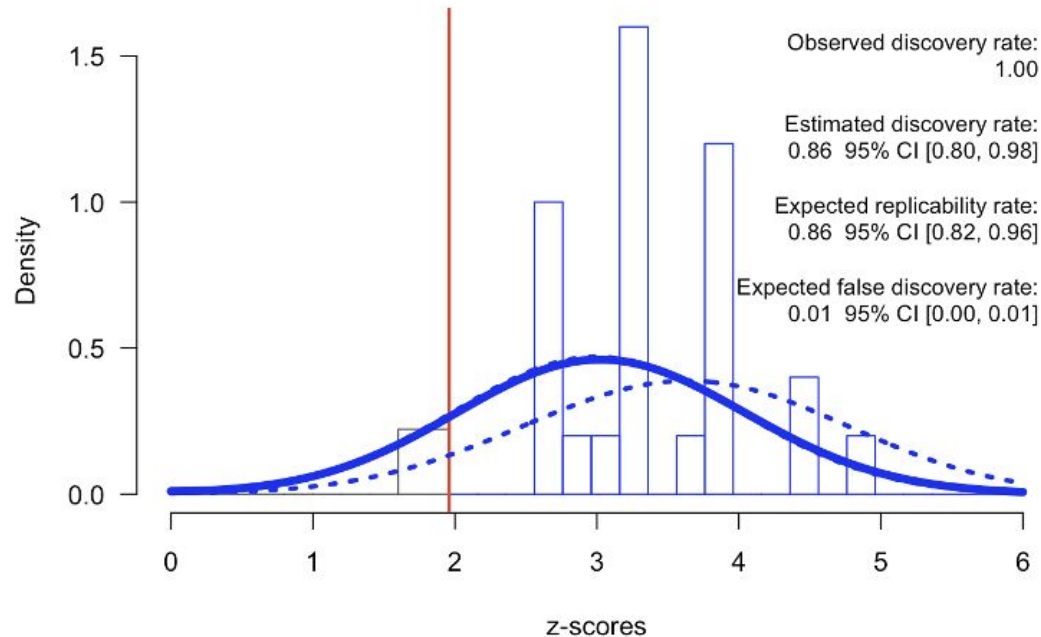


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27/27 of studies included in the analysis

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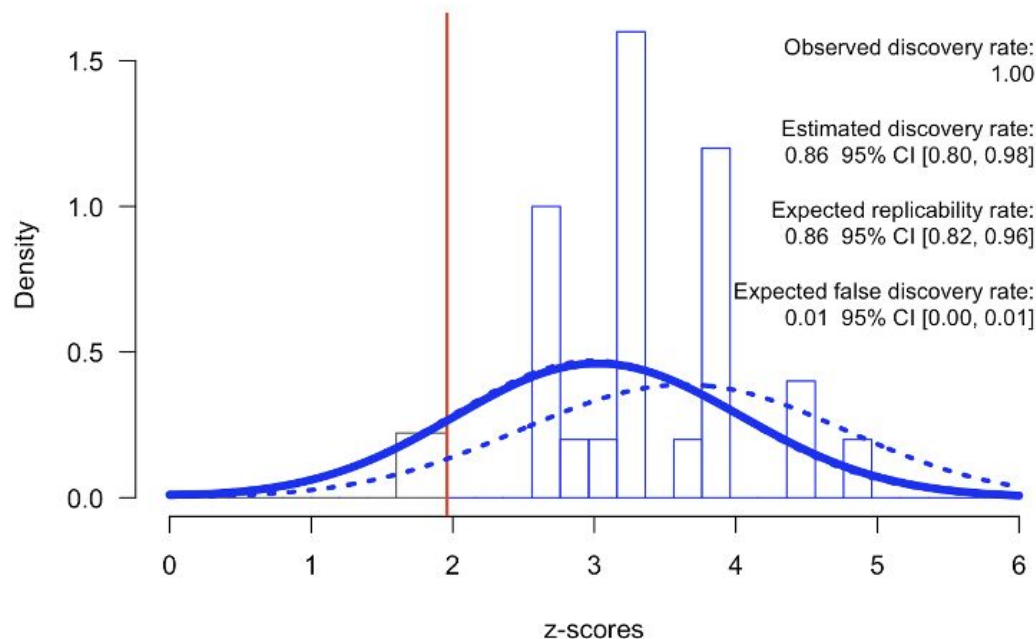


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Solid blue curve $\rightarrow$ The **solid blue curve** is the Z-curve 2.0 model fit to that significant region. What underlying distribution of effect sizes would look like given the *observed* significant z-values

Results and conclusion: Z-curve

Z-curve from uploaded FFA extraction sheet



Red Vertical Line $\rightarrow z = 1.96$

dashed blue curve. It's the model's best guess at what the full distribution of z-values would look like *if every test that was ever run in the field had been reported*

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Expected False Discovery Rate (EFDR)	0.009	[0.001, 0.013]

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The FFA face-selectivity literature looks **credible**

Limitations and Conclusion

- We are in a low-sample regime ($n=27$) reducing confidence in our results
- Some of the literature relating to the FFA is old and does not adhere to modern best practices regarding statistical reporting
- Despite these limitations we still identify genuine large effects with high replicability with no evidence of p-hacking